

IMDEA FINAL PRESENTATION SCRIPT

TITLE SLIDE

PERSONAL INTRO SLIDE

Before we begin...

To reintroduce myself, my name is Ottavia. I just finished my second year at MIT, and I am studying Mechanical Engineering with a focus on Computer Science (Learning Machines and Physical Systems).

I am participating in an international program, called MISTI, where students can work and live abroad over the summer. I came to Madrid in hopes of an interesting work experience, a cultural immersion, and an opportunity to improve my Spanish skills.

Upon arriving to IMDEA, I was tasked with a simulation project. This is what I am here to go over with you all today: an overview of my work over the past two months.

OVERVIEW SLIDE

TES PORTION OVERVIEW SLIDE

WHY TES?

The modern need for energy leads to both climate change and supply & demand concerns, and solar thermal systems can address the mismatch between the intermittent heat supply and the consistent heat demand.

TES is a system where the heat, produced when energy is available, is stored until it is required. It has a high degree of flexibility and energy efficiency.

TES TYPES

TES systems are generally split into three types.

In sensible TES, heat is stored by increasing the temperature of the storage material. This is a durable and commercially available system.

Latent TES systems rely on phase change enthalpies at constant temperature. Heat is stored when the material changes from more to less ordered (e.g., solid to liquid), and the system has higher storage density than sensible TES.

In Thermochemical TES, systems store energy as heat of a chemical reaction in reversible chemical processes. It has the highest storage density, but it is new and still developing.

GREEN HYDROGEN

To provide a bit of context, here is the full project, Green Hydrogen. The TES, here in orange, is provided with temperatures that induce its storage process. It uses PCM capsules to release hot steam, which will then enter the electrolysis process to create hydrogen. The storage system itself seeks to stabilize the generation of steam such that it remains constant.

Latent TES proves to be the ideal solution when we consider its ability to provide stable temperatures and to deal with the variability of solar power. Its efficiency, as we saw in the previous slide, allows it to store a lot of heat during its phase change, without having to change temperature. Thus, it overpowers sensible TES and is more practical than thermochemical TES.

Now, my work here will be to figure out how to decide, size, and organize the system in order to best serve the green hydrogen project.

LATENT TES PORTION OVERVIEW SLIDE

So, now, let's dive into Latent TES.

LTES AND ITS PARTS

The high storage density of latent TES is efficient and can solve the issue of limited space, but its main limitation is its storage material.

This Phase Change Material (PCM) must meet the specifications of the application, which can include parameters like:

- Appropriate melting temperatures
- Large melting enthalpy
- & being stable and chemically inert

For our purposes, we are simulating only one capsule and not the entire tank because of their relative computational complexity. One capsule is much more feasible to model, and its physics can then be applied to represent the tank.

PCM CAPSULES

Some types of PCMs include:

Organic PCMs: have carbon atoms in their structure.

Salt hydrates: formed by water absorption of anhydrous salt.

& Eutectic PCMs, which are mixtures with a particular composition of materials that have the lowest melting point when combined.

These are generally options that can be taken advantage of in different temperature ranges, depending on the application.

ENHANCEMENTS

The biggest challenge of a PCM is to address its low thermal conductivity, essentially with the objective of maximizing heat transfer. This can be assuaged with heat transfer enhancement methods, which are strategies that involve incorporating materials with high thermal conductivity into the capsules.

One important example is an encapsulated system, where the PCM is stored within a small container. They are leak-proof solutions that are very versatile and that usually include a void fraction to allow for thermal expansion during the melting process.

SPHERICAL, CYLINDRICAL, & MICRO CAPSULES

These are all options for storage in encapsulated systems, and according to literature:

Spherical capsules provide closer packing, thus providing greater contact surface and better convective heat transfer.

Relative to spherical, cylindrical capsules have poorer convective heat transfer to the capsule's outer shell, but they have benefits such as production feasibility.

Micro capsules form a slurry (semi-liquid mix) when introduced to a liquid, increasing the natural convective heat transfer within the container.

Overall, the ideal capsule is the one that allows for the highest heat transfer. Notwithstanding the discussions in literature, we will still analyze options for geometry.

OUR CHOICES

We chose Latent TES, since it has greater energy density than sensible systems, and, most importantly, the system can provide steam at constant stable temperatures.

An encapsulated system was also decided on, as it is geometrically versatile and provides higher heat transfer surface area.

As for the PCM, we are going to investigate Urea because it has a suitable melting temperature for our application, but, since there is limited information on its properties and CFD results in literature, we also decided to analyze molten salt, which has similar dimensionless numbers and has very well-established results in literature that can help us validate our model.

As for geometry, we will begin with spherical capsules to (again) compare with results in literature, but we seek to analyze the effect that geometry has on the model as well.

[Urea Prandtl number = 11.6]

PROCEDURE

So, now what?

To make conclusions on our choices, we must develop CFD simulations to model how the PCM capsules work. Different versions must exist for the various methods, materials, and geometries, to see the difference among these options.

After these simulations come alive, we must experiment with real life materials to actually validate the models.

Only then can we draw conclusions and make generalizations to apply the simulations to the Green Hydrogen project.

CFD MODELING PORTION OVERVIEW SLIDE

COMPUTATIONAL FLUID DYNAMICS

Computational Fluid Dynamics (CFD) is the analysis of systems involving fluid flow, heat transfer, and associated phenomena. Based on the principles of fluid mechanics, models simulate flow and the interaction of liquids and gases.

Three fundamental equations governing fluid motion:

- Conservation of mass
- Conservation of momentum
- Conservation of energy

CFD CHARACTERISTICS

CFD analysis consists of three main steps: first, pre-processing (where we set the geometry and meshing), then processing (where the software solves the governing equations in the discretized geometry), and, lastly, post-processing (where the software processes these results).

The solver uses one of three distinct numerical solution techniques: finite difference, finite element, or spectral methods.

Working simulation results are characterized by convergence and grid independence, &

The accuracy and cost of any CFD solution is based on the number of cells chosen in that grid. Optimal meshes are finer in areas with large variations and coarser in areas with less change.

[Spectral methods are essentially discretization methods for the approximate solution of partial-differential equations expressed in a weak form, based on high-order Lagrangian interpolants used in conjunction with particular quadrature rules.]

ENTHALPY POROSITY METHOD

We are using the enthalpy porosity method for our models. This fixed mesh approach trumps using a separate mesh for solid and liquid phases due to minimized computational complexity. By fixing the mesh, we are making the solid state a porous medium with 0 apparent porosity. This method is already implemented in ANSYS, a different CFD software, and we are just adding it to our COMSOL simulation.

Here are the fundamental equations that simulate fluid flow and heat transfer, through the conservation of mass, energy, and momentum.

[The specific heat, C_p , is the energy needed to increase temperature by one Kelvin. Its function is given by the average of its liquid and solid densities, added to its latent heat multiplied by its melt fraction, which represents the distribution of latent heat over the mushy zone temperature interval.]

Next slide

This is the momentum source term used in the momentum equation. The first term, ρg , is always included in the momentum while the second term serves to introduce the phase change effect on convection.

The porosity and melt fraction range from 0 to 1, or liquid to solid.

CFD SIMULATION LITERATURE

The agreement between numerical and experimental results of a single capsule melting process for molten salt, and other representative materials, has been well-established in literature, as concluded in the following research papers, and others, for both spherical and cylindrical capsules.

COMSOL SOFTWARE

COMSOL is a simulation software that uses the finite element method and is relatively intuitive and user-friendly. It has great access to physics, and it is open to user input. However, the simple user interface makes it difficult to debug computational problems, as we have seen many times.

In our case, we are using the COMSOL CFD software to analyze and model fluid flow, heat transfer, and melt fraction. As for results, trends of density, pressure, temperature, and more, can be plotted and animated.

COMSOL PCM SIMULATIONS PORTION OVERVIEW SLIDE

MODEL

The fundamental geometry we are modeling to start off with is made using a semicircle in the 2D axisymmetric mode, using a time-dependent laminar flow study. The initial temperature and boundary wall temperature are the two most important conditions. Later, we will play around with these conditions and the geometry.

Some of our simplifications include using laminar flow, ignoring boundary material and liquid to solid thermal expansion, and assuming constant boundary temperature and solid density.

COMSOL SIMULATION PATH

The COMSOL gradual simulation path that we created included:

- Sphere with water
- Some individual exploration
- Introduction to phase change physics and making new materials
- Molten Salt Spherical PCM Capsule MELTING
- Molten Salt Spherical PCM Capsule SOLIDIFICATION
- And applying melting and solidification simulations to other materials, such as Urea, Paraffin Wax, and water, and other geometries, such as cylinders, and squares.

Our end goal was for the simulation to accurately represent the fluid flow, heat transfer, and melt fraction of the capsules, with varying materials and geometries.

1. SPHERE WITH WATER

Simulations started off with introducing geometries, laminar fluid flow, and heat transfer, simply using water. Here are these initial, very preliminary results.

2. INDIVIDUAL EXPLORATION

My individual exploration involved playing around with the geometries, materials, and results on the COMSOL software, to get a feel for the platform. Here is a cylindrical capsule with Mercury, (*next slide*) a spherical one with gasoline, and a semi-hollow cylinder with hydrogen.

3.1 PHASE CHANGE PHYSICS

The simulations then began to include phase change physics and features. We changed the material property equations to account for phase change and the mushy zone, specifically for ice and water in

this case. These results are still part of my COMSOL exploration and familiarization, as various features, like momentum, are not yet included.

3.2 NEW MATERIAL

We then focused on creating a new material base within COMSOL to represent the Molten Salt of our first model. I implemented new equations for the PCM properties and created a mushy zone interval for density using linear interpolation. We then edited most of these plots, but this base would allow us to create blank materials for Urea and Paraffin Wax later in the project.

4. SODIUM NITRATE SPHERICAL PCM CAPSULES: MELTING

Then, a representative simulation for Molten Salt was created. This simulation went through many versions and suffered many modifications. The process began with creating the basic spherical capsule representation using the new Sodium Nitrate material that we created in COMSOL.

4. VERSION 2

Then, modified equations for a volume force, the source term, were included in the simulation to represent the solid-state phase change.

The source term equation used can be seen on the slides, whose components are negative due to resistance to flow. And these are the initial values for the melting front morphology constant, C or A-mushy, and the computational constant, epsilon.

4. VERSIONS 3, 4, 5 (OPTIMIZATIONS)

Then, in versions 3, 4, and 5, there were many optimizations.

The Dynamic Viscosity (μ) property plot was modified to account for different phases of the material. The liquid phase viscosity is 0.003 [Pa*s], while the solid phase viscosity was set to an arbitrarily large value of 1000 [Pa*s].

Initially, we allowed COMSOL to choose appropriate timesteps for each simulation based on the physics, but, since the model had new and user-input physics involved, we decided to use a fixed timestep.

Other optimizations integrated included the tolerance factor, the maximum number of iterations, the mesh, and the mushy zone. These were all continuously adjusted between each version.

Most simulations took a long time to run, upwards of 48 hours, so we got very good at waiting.

[The default value of the relative tolerance is 0.01 and the Tolerance Factor multiplies this number, so setting the Tolerance factor to 0.1 in this case would result in a relative tolerance of 0.001 used for the nonlinear problem at each timestep. Thus, as you tighten the time-dependent solver relative tolerance setting, you can usually relax the tolerance factor.]

4. VERSION 5: TEMPERATURE & MELT FRACTION

Here we can see the 2D and 3D temperature and melt fraction trends for this version. The two results are generally consistent with each other.

4. VERSION 5: 2D FLUID FLOW

Next, we can see the capsule's fluid flow, as a rainbow and arrow surface. We noticed some inaccuracies with this flow that will be addressed and corrected later in this presentation.

4. VERSION 5: COMBINED

Lastly, here is a combination of this version's most important results. We have melt fraction as a greyscale surface, temperature as a contour, and fluid flow as an arrow surface.

4. VERSION 6: RECIRCULATION

As evident in the previous fluid flow results, we were not seeing any recirculation trends in the capsule, which we expected due to the material's density properties and based on pertinent literature.

We thus updated the density piecewise plot such that liquid density was represented by a temperature dependent function. This allowed us to see recirculation flow trends in the newer results.

We also updated the melting front morphology constant, to a more accurate value of 10^4 , which allowed us to see the true heightened damping effects and slower melting time, seeing as the momentum source was largely scaled.

New slide

Here we can see the overall results of this recirculation version: the temperature, and the melt fraction and fluid flow. The fluid flow moves up the side of the capsule and recirculates downwards towards the solid phase.

5. SODIUM NITRATE SPHERICAL PCM CAPSULES: SOLIDIFICATION

We then seek to model the opposite process, transforming from liquid to solid. Thus, we switch the initial and boundary temperatures, starting at a high temperature (begins as liquid) with a low wall boundary temperature (becomes a solid).

5. SOLIDIFICATION VERSION

Here we can see 2D temperature and melt fraction animations for the Molten Salt solidification process. We noticed an inaccuracy with this pressure point right here, but at this point we still had not made any recirculation modifications.

Indeed, for this model, we updated the liquid density and melting front morphology constant, to get accurate recirculation in the capsule. However, we were not able to get these results to converge in time for today, so they are not final nor completely accurate.

6. MOLTEN SALT MELTING: DIFFERENT GEOMETRIES

We also created models with two different types of geometry, one with a cylindrical capsule, and one of a square.

Here we can see the results for the Molten Salt Cylindrical model. Later, we will compare these to its spherical version.

We also began working on modeling the square version, but with such strong boundaries, we have yet to fine tune its mesh to achieve convergence.

6. UREA: SPHERICAL CAPSULE

The last two simulations we ran this week were two simulations using a rendering of the Urea material, which is ideally the material that will be used in the final capsule. Here are the results for the Urea spherical capsule:

We also ran a simulation for a Urea cylindrical model but that actually crashed this morning.

MATERIAL COMPARISONS

This plot compares the melt fraction rates by material, of the Sodium Nitrate Molten Salt and the Urea models. Over a span of 240 seconds, the two materials have pretty similar melt fraction rates, but the Urea seems to always be slightly ahead, with a higher final melt fraction. This trend seems as though it would continue over time, but we would need a longer simulation to know for sure.

GEOMETRY COMPARISONS: MOLTEN SALT

This plot compares the Molten Salt melt fraction rates by geometry, of the spherical and cylindrical capsules. Over a span of 660 seconds, the two geometries have very similar melt fraction rates. The spherical capsule has a slightly higher liquid fraction at the end, as predicted in literature, but we would need to analyze a longer timeframe to make any conclusions.

POTENTIAL CONCERNS AND UNCERTAINTIES

There are various assumptions or features that may be cause for uncertainty and should be further explored in the future. These may include:

- The value of C , which can be deduced from conducting the experiment
- The phase change hysteresis phenomenon
- The boundary layers, mesh, and timesteps
- The pressure points, viscosity values, and mushy zone temperature interval

CFD MODEL VALIDATION EXPERIMENT PORTION OVERVIEW SLIDE

EXPERIMENT PLAN/METHOD

The main idea of this validation experiment was to use Paraffin Wax (PW) as a representation for the Urea PCM capsule because it has a lower melting point. This meant that we would not need a high temperature oven, making the validation a lot easier. We would then validate our CFD results using the experiment, potentially scale these measurements according to our calculated dimensionless numbers, and, lastly, conclude that the model can be used for other materials and conditions.

EXPERIMENT DESIGN

Our method plan to set up the experiment was to use a water tank, which includes a temperature pump to heat the water, and a built-in fan to make sure the water flow remains constant.

Measurement methods:

- Use thermocouples to measure temperature within and outside the capsule, for data and to confirm constant boundary temperature
- Use a camera to view and measure the melt fraction

With these values, we would compare the PW numerical and experimental results and, if accurate and congruent, deduce that we can apply the simulation model to Urea PCM.

LAB MATERIALS

These are the actual lab materials that we planned on using for the experiment. These beakers, these thermocouples, and this hot plate instead of the water bath.

DIMENSIONLESS NUMBERS

To make an accurate hypothesis on the behavior of these materials, our top priority was to make sure the experiment's and the final material's dimensionless numbers aligned with one another. We wanted to choose a paraffin wax that had the ideal properties, such that the Stefan, Prandtl, and Rayleigh numbers were as similar as possible.

However, this was not easy to find, and we thus decided to validate the models by creating a simulation for Paraffin wax on its own and comparing it to our experimental results, instead of using dimensionless numbers.

PARAFFIN WAX AND WATER SIMULATIONS

For the experiment, we wanted to simulate a regular paraffin wax model as well as one composed of equal solid and liquid phase densities, meaning that when the PCM solidifies, it won't fall to the bottom of the capsule. This model represents the presence of thermocouples during the experiment, since the solid will be attached to the rod and won't move.

We also wanted to conduct this constrained model on the phase change of ice and water. Since ice has a lower density than water, its unconstrained melting would not be applicable, yet, when the ice is constrained to the rod and densities do not cause differences, water serves as an uncanny representation for Urea, with a very similar Prandtl number.

NEXT STEPS

Time willing, I would have liked to finish the procedure, so here are the next steps at this point in the process.

First off, I would complete the simulations pertaining to the experiments, meaning the constrained and unconstrained paraffin wax models and the constrained water model.

Then, I would complete the experiment plan set forth in the previous slides. Using these measurements, we could compare the dimensionless numbers and numerical vs experimental results to see if it would be possible to validate the simulation method.

If this were the case, I would then move on to complete simulations of all intended materials and geometries. With more time, I would also include an air gap in the capsule, to account for thermal expansion, and a solid material boundary outside the capsule.

Lastly, going forward, other more applicable experimental simulations on the new choices for the capsule should be conducted, tested, and validated.

CONCLUSIONS

Over these past two months, we were able to make conclusions on the effect of two materials and two geometries on the temperature, flow, and, most importantly, the melt fraction of a PCM capsule.

But, overall, throughout *my* time spent working on CFD models, I have learned to use the COMSOL software to represent different geometries, materials, and features of systems.

Although there are endless applications of these models, I am now comfortable in my abilities to explore the program and teach myself additional physics through online resources.

MADRID PERSONAL EXPERIENCE PORTION OVERVIEW SLIDE

MADRID EXPERIENCE

The past two months in Madrid have been unlike any other experience. I had to be the most independent I've ever been, and I felt as though I grew up a lot this summer.

I was able to travel to many places over the weekends, which I really enjoyed. Within Madrid, I was stunned by the architecture, the food, the happy atmosphere of the streets, and the crazy heat.

I also found a beach volleyball team, and I was so happy to be able to keep playing even in across the world.

As for the Spanish language, I think my skills have improved. Although I still often get confused, I am more confident in my ability to communicate with Spaniards. I hope to continue speaking Spanish, and I especially hope to come back to Madrid soon.

THANK YOU PORTION OVERVIEW SLIDE

THANK YOU!

Thank you all!

A special shoutout to Jose for getting me here through MISTI and allowing me to have this experience, I am extremely grateful.

And a special shout out to Germilly for all his help. He was always available, ready for a challenge, and very supportive throughout the whole process.

I wish you all the best of luck, it has been an honor to work with you and I hope to see you again soon!

WORKS CITED

QUESTIONS?